

Longest Common Subsequence

atlantis

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• Introduction to LCS (Longest common sequence) =

• Sub string $\rightarrow$ RAVINDRA
(RAVI, AVI, RAV)

Sub string should be contiguous.

• Subsequence $\rightarrow$ RAVINDRA
1 2 3 4 5 6 7 8

{1, 2, 3, 4} RAVI

{1, 3, 5, 8} RVNR

{2, 7} AR

{1, 2, 5, 6, 8} RANDA

{ }

Sub sequence ~~should~~ ^{could} not be contiguous.
should

$\rightarrow$ string with length 'm' $\rightarrow$ 2^m subsequence possible.

$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \rightarrow 2^8$
RAVINDRA

there are 4 molecule (A, C, G, T) using which DNA made off.

$D_1 = \text{A G C C T C A G T}$

$D_2 = \text{G C C C T}$

Similarity of D_1 and D_2 depends on Longest Common sequence.

(AAGTGA) $\begin{bmatrix} D_1 \\ D_2 \\ D_3 \end{bmatrix}$ (ACGT) = LCS(3)
LCS(5) (ACGT) = LCS(4)

here D_1 and D_3 very similar compare to any other pair.

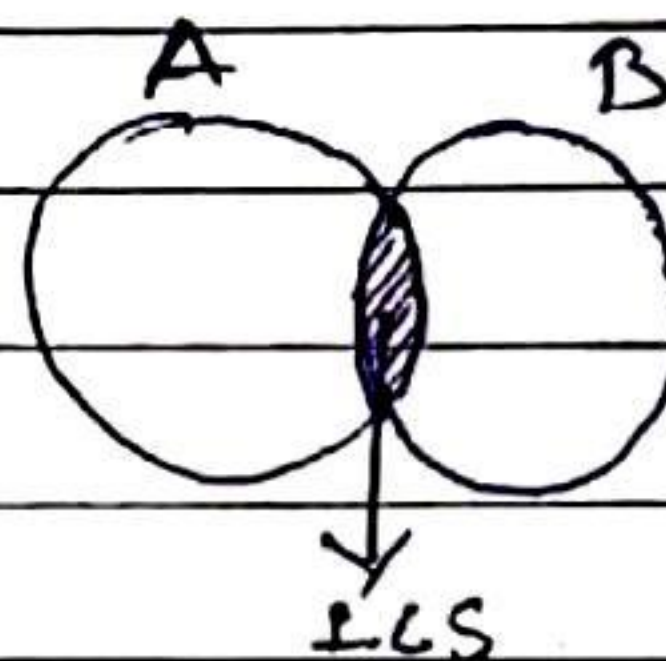
- (i) string - {RAVINDRA} $\rightarrow$ string length $m \rightarrow$ no. of Subsequence Possible 2^m
- (ii) string - {AJAY}
- here LCS is = {A, A} (Longest Common Subsequence)

Algorithm =

- I. Find all subsequence of 'A' $- O(2^m)$
- II. Find for each subsequence whether it is a subsequence of 'B'. $- O(n2^m)$
- III. Find the longest among common subsequence. $- O(2^m)$

A = RAVINDRA

B = AJAY



$\rightarrow$ here Time complexity = $O(n2^m)$.

• Optimal substructure and Recursive equations =

Optimal substructure

$$D_1 = x_1 x_2 x_3 \dots x_n$$

$$D_2 = y_1 y_2 y_3 \dots y_m$$

$$x_1 x_2 x_3 \dots x_i$$

$$y_1 y_2 y_3 \dots y_j$$

Recursive equation -

$$c[i, j] = \begin{cases} 0, & i=0 \text{ or } j=0 \\ 1 + c[i-1, j-1], & i, j > 0 \text{ and } x_i = y_j \\ \max(c[i-1, j], c[i, j-1]), & i, j > 0 \text{ and } x_i \neq y_j \end{cases}$$

Recursive tree and unique sub problems -**ex 8**

$$X = \{A, A, A, A\}$$

$$Y = \{A, A, A, A\}$$

best case input.



$$C(4,4)$$

$$1 + C(3,3)$$

$$1 + C(2,2)$$

$$1 + C(1,1)$$

$$1 + C(0,0)$$

$$\text{So, LCS} = 4$$

(5-function call)

function call - $O(n)$ time complexity - $O(n)$ space - $O(n)$ **ex 9**

$$X = \{A, A, A, A\}$$

$$Y = \{B, B, B, B\}$$

Worst case i/p.



$$C[4,4]$$

$$C(4,3)$$

$$C(3,4)$$

$$C(3,3)$$

$$C(4,2)$$

$$C(2,4)$$

$$C(3,3)$$

$$C(2,3)$$

$$C(3,2)$$

$$C(1,3)$$

$$C(2,2)$$

$$C(0,3)$$

$$C(1,2)$$

$$C(0,4)$$

$$C(1,1)$$

$$C(0,1)$$

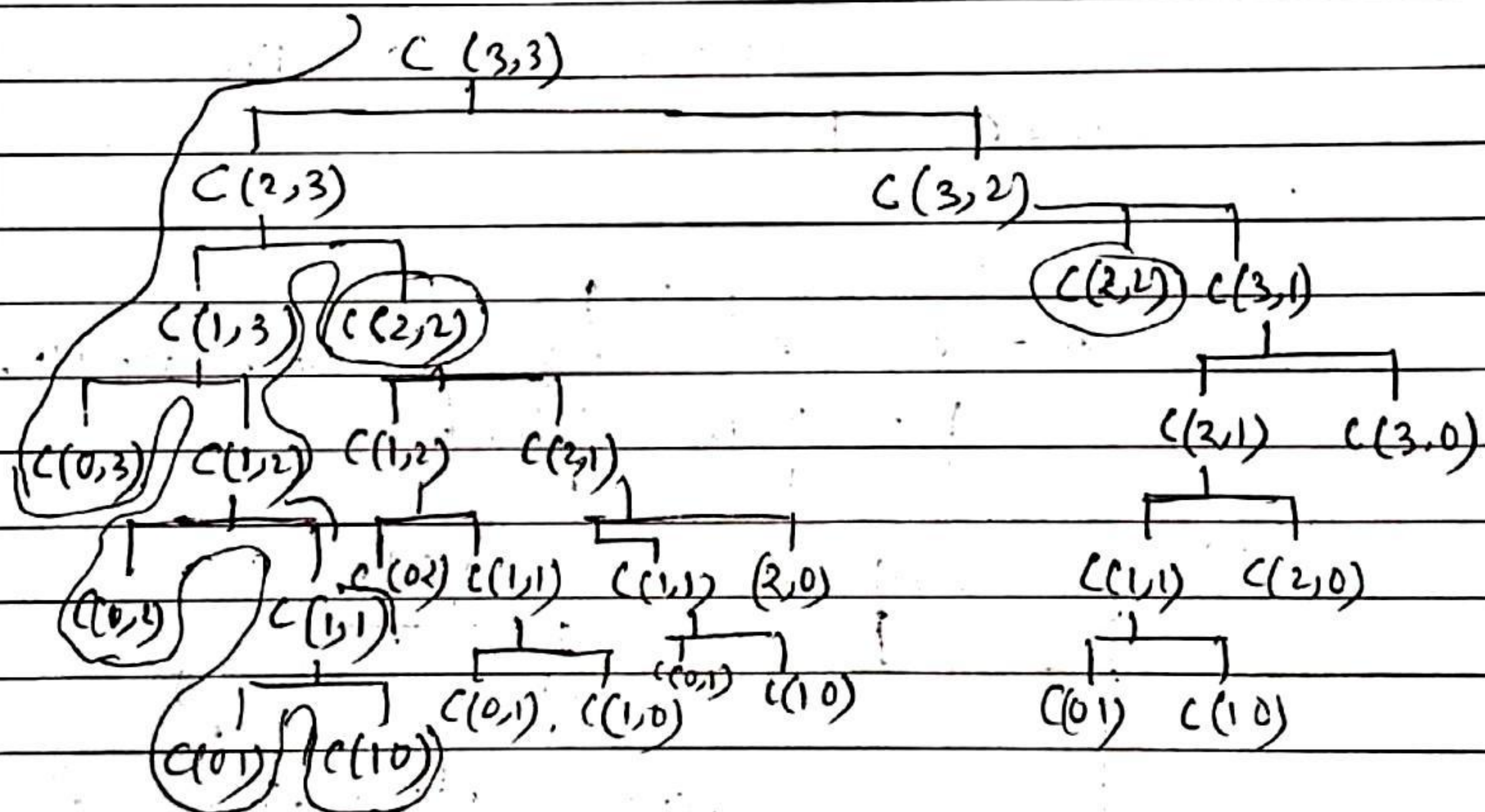
$$C(1,0)$$

$C[n, m]$

Worst case

 $C[n-1, m]$

no common subsequence

 $C[n-1, m-1]$ depth of tree = $O(n+m)$ $C[n-2, m-1]$ no. of node = $O(2^{n+m})$ $C[n-2, m-2]$ time complexity = $O(2^{n+m})$ $C[0, 1]$ $C[0, 0]$ ex: unique sub problems =

$C(3,3)$	$(3,3)$	$(3,2)$	$(3,1)$	} - 9 subproblem
$C(2,3)$	$(2,3)$	$(2,2)$	$(2,1)$	
$C(1,3)$	$(1,3)$	$(1,2)$	$(1,1)$	
$C(0,3)$	$(0,0)$	$(2,0)$	$(3,0)$	} - 16 subproblem
$C(1,2)$	$(0,1)$	$(0,2)$	$(0,3)$	
$C(0,2)$	$(0,0)$			
$C(1,1)$				
$C(0,1)$				
$C(1,0)$				

	0	1	2	3
0	00	01	02	03
1	10	11	12	13
2	20	21	22	23
3	30	31	32	33

no. of unique function = m^n
 call - $m \times n$

Time complexity = $O(mn)$

Space complexity = $O(mn)$

[ex^o]

~~$X = A, A, B, 1$~~

$X = (AAB)$

~~$X = (AAB)B$~~

$Y = (ACA)$

~~$Y = (CBDB)$~~

sub
 Find longest common sequence
 using table -

→ $Y = A C A$

	0	1	2	3
X	0	0	0	0
A 1	0	1	1	1
A 2	0	1	1	2
B 3	0	1	1	2

$\begin{cases} \text{match} = 1 + C(i-1, j-1) \\ \text{no match} = \max(C(i-1, j), C(i, j-1)) \end{cases}$

$i+0 = 1$

$1+1 = 2$

$i \rightarrow \text{LCS} (AA)$

Exo

Two DNA given,

 $X = \{A, B, C, B, D, A, B\}$ $Y = \{B, D, C, A, B, A\}$

find out Longest common subsequence.

		0	1	2	3	4	5	6	
	Y	B	D	C	A	B	A		
0 X	0	0	0	0	0	0	0		
1 A	0	0	0	0	1	1	1		• when match $1 + C(i-1, j-1)$
2 B	0	1	1	1	1	2	2		
3 C	0	1	1	2	2	2	2		• when no match.
4 B	0	1	1	2	2	3	3		
5 D	0	1	2	2	2	3	3		$\leftarrow C(i, j-1)$
6 A	0	1	2	2	3	3	4		$\leftarrow C(i, j)$
7 B	0	1	2	2	3	4	4		$\max(C[i-1, j], C[i, j-1])$

- 1) BCBA $Y = 1, 2, 4, 5$ $X = 2, 3, 6, 7$ B C A B present
- 2) BDAB $Y = 1, 2, 4, 5$ $X = 2, 5, 6, 7$ B D A B
- 3) BCAB $Y = 1, 2, 4, 5$ $X = 2, 5, 6, 7$ B C A B
- Longest common subsequence

here, LCS = 4

no. of subsequences = 3

• Bottom up Dynamic programming algo-

$LCS(X, Y)$

{

$m = X.length$

$n = Y.length$

let $C[1 \dots m, 1 \dots n]$ be a new table.

for $i = 1$ to m

$C[i, 0] = 0$

for $j = 0$ to n

$C[0, j] = 0$

for $i = 1$ to m

for $j = 1$ to n

if $x_i == y_j$

$C[i, j] = C[i-1, j-1] + 1$

else

$C[i, j] = \max(C[i-1, j], C[i, j-1])$

return C .

}

Time Complexity = $O(mn)$

space Complexity = $O(mn)$

9-2019

Question]-①

consider two strings $A = "qpqr r"$ and $B = "pqprq -rp"$.

Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the no. of such LCS b/w A and B . Then $x + 10y = ?$

$\rightarrow A = q p q r r$

$B = p q p r q r p$

	0	1	2	3	4	5
0 B	A-q	p	q	r	r	
1 P	0	0	1	1	1	1
2 q	0	1	1	2	2	2
3 P	0	1	2	2	2	2
4 r	0	1	2	2	3	3
5 q	0	1	2	3	3	3
6 r	0	1	2	3	4	4
7 P	0	1	2	3	4	4

$$X=4$$

$$Y=3$$

(1) p q r r (4) q p q r

(2) q p q r

(3) q p r r

$$\text{So, } x + 10y$$

$$= 4 + 10 \times 3 = \cancel{44} / 30 + 4 \Rightarrow \boxed{34} \checkmark$$

9-2009

Question - (2)

We are given two sequences $x[m]$ and $y[n]$ of lengths m and n , respectively, with indices of x and y starting from 0.

Q1 find the length of the longest common subsequence (LCS) of $x[m]$ and $y[n]$ as $l(m, n)$, where an incomplete recursive definition for the function $l(i, j)$ to compute the length of the LCS of $x[m]$ and $y[n]$ is given below -

$$\begin{aligned}
 l(i, j) &= 0, \text{ if either } i=0 \text{ or } j=0 \\
 &= \text{expr1}, \text{ if } i, j > 0 \text{ and } x[i-1] = y[j-1] \\
 &= \text{expr2}, \text{ if } i, j > 0 \text{ and } x[i-1] \neq y[j-1]
 \end{aligned}$$

Which of the following options is correct -

~~(a)~~ $\text{expr1} \equiv 1 + l(i-1, j)$

(b) $\text{expr1} \equiv l(i, j-1)$

☒ (c) $\text{expr2} \equiv \max(l(i-1, j), l(i, j-1))$

(d) $\text{expr2} \equiv \max(l(i-1, j-1), l(i, j))$

→ $\text{expression 1} \equiv 1 + l(i-1, j-1)$

Q22

$l(i, j)$ is computed by DP (Dynamic programming) using an array $L[M, N]$, where $M = m+1$ and $N = n+1$, such that $L[i, j] = l(i, j)$.

Which of the following statement is TRUE -

~~(a)~~ All elements of L should be initialized to 0.

☒ (b) The value of $L(i, j)$ may be computed in RMO or LMO

~~(c)~~ The values of $L(i, j)$ cannot be computed in either RMO or LMO

☒ (d) $L(p, q)$ needs to be computed before $L(r, s)$ or if either $p < r$ or $p < s$.